



CSE 3313: Computer Architecture

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Spring 2025 Mid Term Question



1. Suppose there are three classes of instructions A, B, and C in a particular instruction set architecture with CPIs 2, 1.5, and 2.2 respectively. The number of instructions(IC) from each class in two separate programs is as follows:

Programs	A	B	C
P1(IC)	100	110	116
P2(IC)	112	100	40

Consider that both programs run on the same device with a clock frequency of 2 GHz. Both P1 and P2 require 450 ns to be completed.

- (i) Find out the affected and unaffected times for both programs. [2]
- (ii) Find the improvement factor for P1 and P2 if we get the total program completion time improved by 3x. [6]
- (iii) If P2 now has the same execution time as P1 what should be the number of instructions (IC) for instruction class C? [2]

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Suppose there are three classes of instructions A, B, and C in a particular instruction set architecture with **CPIs 2, 1.5, and 2.2 respectively.**

[It means each A requires 2 Clock pulse, B require 1.5 clock pulse, C require 2.2 clock pulse]. The number of instructions(IC) from each class in two separate programs is as follows:

Programs A B C

P1(IC) 100 110 116

P2(IC) 112 100 40 **[for P1, A= 100 means there are 100 A instructions in P1]**

Consider that both programs run on the same device with a clock frequency of 2 GHz. **Both P1 and P2 require 450 ns to be completed. { this is the response time}**

(i) Find out the affected and unaffected times for both programs.

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CPU Execution Time = Affected Time

Unaffected Time = Total Response time - affected time

CPU Execution Time = Total Clock Pulse / Clock Frequency

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For P1:

Total cycles = $\{100(2) + 110(1.5) + 116(2.2)\} / 2 \times 10^9 = 310.1 \text{ ns}$

For P2:

Total cycles = $\{112(2) + 100(1.5) + 40(2.2)\} / 2 \times 10^9 = 231 \text{ ns}$

Both take 450 ns.

That means actual execution is longer than base execution → extra overhead.

Affected time = Base execution time (calculated).

Unaffected time = Extra overhead = $(450 - \text{base time})$.

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P1: unaffected = 310.1 ns, affected = $450 - 310.1 = 139.9$ ns

P2: unaffected = 231 ns, affected = $450 - 231 = 219$ ns

P1: Affected = 139.9 ns, Unaffected = 310.1 ns

P2: Affected = 219 ns, Unaffected = 231 ns

1. Suppose there are three classes of instructions A, B, and C in a particular instruction set architecture with CPIs 2, 1.5, and 2.2 respectively. The number of instructions(IC) from each class in two separate programs is as follows:

Programs	A	B	C
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Consider that both programs run on the same device with a clock frequency of 2 GHz. Both P1 and P2 require 450 ns to be completed.

- (i) Find out the affected and unaffected times for both programs. [2]
- (ii) Find the improvement factor for P1 and P2 if we get the total program completion time improved by 3x. [6]

**Improvement factor is always on response time.
Here response time is 450 ns as given in question.**

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Improvement factor with 3× speedup

If improved execution time = $(450 \div 3) = 150$ ns.

$T_{\text{improved}} = 150$ ns

For P1,

$$150 = (310.1 / k) + 139.9$$

$$K = 31$$

The value of K is negative. So this speedup is not possible.

For P2,

$$150 = (231 / k) + 219$$

$$k = 231 / (150 - 219) = -3.348$$

The value of K is negative. So the speedup is not possible.

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(iii)

If P2 has same execution time as P1

We want: P2 execution time = P1 execution time = 310.1 ns (base).

Convert to cycles:

$$310.1 \div 0.5 = 620.2 \text{ cycles.}$$

We already have for P2 (without change):

$$112(2) + 100(1.5) + (C \times 2.2) = 224 + 150 + 2.2C$$

$$C \approx 112$$

Instruction class C should have 112 instructions.

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1. Consider computer A with a CPU speed of 2 GHz. The program P which has number of instruction and CPI as shown in Table 1.

Instruction	Count	CPI
addi	4	2
add	4	1
beq	2	4
bne	1	4
slt	2	3
sll	2	4
and	3	2
j	2	8

Table 1: Program P's number of instructions and CPI in Computer A architecture

Consider that Program P requires 50 ns to be completed Computer A.

- Find the execution time of program P in Computer A. [2]
- Computer B with a CPU speed of 10 GHz has the execution time of the program P which is now 2 times faster than computer A and same instruction set architecture (ISA), calculate the average CPI of computer B. [2]
- As we get the total execution time of the given program for Computer A from question (i), assume that the Arithmetic unit takes 65%, the logical unit takes 25%, and the branch operation takes 10% time of total execution time. Find the improvement factor of the given program if we replace the arithmetic unit with a better Arithmetic unit, which improves total completion time by 2x. [6]

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(i)

Find the execution time of program P in Computer A. [

Total Clock Pulse=

$$(4 \times 2) + (4 \times 1) + (2 \times 4) + (1 \times 4) + (2 \times 3) + (2 \times 4) + (3 \times 2) + (2 \times 8) = 60 \text{ cycles}$$

$$\text{Clock Rate} = 2 \times 10^9 \text{ Hz}$$

$$\text{CPU Time} = 60 / 2 \times 10^9 = 30 \times 10^{-9} \text{ s} = 30 \text{ ns}$$

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(ii)

Execution time on B = $30 \text{ ns} / 2 = 15 \text{ ns}$

Clock Rate = $10 \times 10^9 \text{ Hz}$

Total Instructions = 20

Total Clock Pulse = Execution time x Clock Rate = $15 \text{ ns} \times 10 \times 10^9 \text{ Hz} = 150$

Average CPI = Total Clock Pulse / Total Instruction = $150 / 20 = 7.5$

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(iii)

Given total completion time for program P on A is 50 ns and execution time is 30ns from question (i)

Arithmetic: 65% of 30 ns = 19.5 ns

Logical: 25% of 30 ns = 7.5 ns

Branch: 10% of 30 ns = 3 ns

After improving total completion time by 2x:

Improved time = $50 / 2 = 25$ ns

Affected time = 19.5ns

Unaffected time = $7.5 + 3 + 20(\text{system}) = 30.5$

so, $25 = 30.5 + 19.5/n$

2x improvement not possible, even if we replace the arithmetic unit with a better one

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1. A program P runs on computer A with a clock frequency of 2 GHz. The instructions of the program are classified with the following distribution:

Instruction Type	R Type	I Type	J Type
Instruction Count	10	5	4
Cycles Per Instruction	3	6	2

Table 1: Program P Description

- (a) If program P completes its entire execution after 50 ns, find out the affected and unaffected time. [4]
- (b) A second computer B, with the same Instruction Set Architecture as that of A, wants to run program P in half of the execution time as that of A. Calculate the required clock frequency of computer B. [3]
- (c) Find out the improvement factor if the execution of R type instructions can be improved by 2 times. [3]

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a) Total Cycles= $(10 \times 3) + (5 \times 6) + (4 \times 2) = 30 + 30 + 8 = 68$
Execution Time at 2 GHz:

Time= Clock Rate / Total Cycles = $68/2 = 34$ ns

Affected Time / Execution Time = 34 ns

Unaffected Time= $50 - 34 = 16$ ns

b) Half of Execution Time of A = $34/2 = 17$ ns

Total Clock Cycle = 68 [same ISA]

Clock Rate= $68/17 = 4$ GHz

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c) Clock Cycle of R type = $10 \times 3 = 30$
Clock Rate = 2 Ghz

Time for R type = $30 / 2 = 15$ ns

If Time for R type improved 2 times , new time for R type = $15/2 = 7.5$ ns

Time for I type = $5 \times 6 / 2 = 15$ ns

Time for J type = $4 \times 2 / 2 = 4$ ns

New Execution Time = $7.5 + 15 + 4 = 26.5$ ns

T_{improved} = 26.5 ns

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Affected Time= 15 ns

Unaffected Time = 50 - 15 = 35 ns

Using Amdahl's law

$$26.5 = 15/n + 35$$

$$n = 15 / (26.5 - 35) = -1.765$$



Thank you!